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Evidence for enrichment of subcontinental lithospheric mantle from Paleoproterozoic intracratonic magmas: Geochemistry and U–Pb geochronology of Martin Group igneous rocks, western Rae Craton, Canada

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ABSTRACT

The Paleoproterozoic Martin Group, located in the Rae Craton of northwestern Saskatchewan, is a sequence of intercalated siliciclastic deposits and mafic igneous rocks that were deposited in a fault-controlled intracratonic basin. Based on field relationships, mafic volcanic rocks and sills of the Martin Group have historically been viewed as genetically related to a suite of regionally extensive mafic dikes. In this study, the geochemical characteristics of the three mafic suites (volcanic rocks, sills, and dikes) were used to interpret their respective magmatic histories, to assess the possibility of a genetic relationship between the suites, and to deduce the nature of the magma source. Geochemically, these rocks range from subalkaline (tholeiitic) basalts to alkali basalts or trachyandesites and contain trace element compositions consistent with emplacement in an intracontinental setting. All three suites are highly enriched in light rare earth elements but have anomalously low concentrations of high field strength elements (e.g. Nb, Ta, Hf, Zr) relative to other trace elements. The trace element compositions suggest derivation of all three suites from a common subcontinental lithospheric mantle source that had experienced a complex enrichment history, including modification of depleted mantle by ancient subduction-related components and by small degree asthenospheric melts. A common, enriched source for these magmas is confirmed by their identical initial Nd isotopic compositions (ε_{Nd} at the time of emplacement of −3.7 to −5.3), likely indicative of input of isotopically evolved crust to the source region during the subduction-related enrichment. The overall similarity of chemical compositions for the three suites indicates that regional emplacement of the mafic dikes and (localized?) magmatism within the Martin Group was essentially part of the same mafic magmatic event in the area, though slight intersuite compositional variations might reflect differences in magma evolution histories. A U–Pb baddeleyite date from one dike broadly dates this magmatic event at ca. 1818 Ma, though it must have occurred as a series of magmatic pulses that alternated with periods of sedimentary deposition within the Martin basin. The dike magmas infiltrated pre-existing fractures, whereas the intrabasinal suites might have utilized major faults associated with basin genesis (e.g. the Black Bay fault) as magma conduits.

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1. Introduction

The Rae Craton and adjacent Hearne Craton to the east (Fig. 1) form the Archean nucleus of the Churchill Province, one of multiple Archean provinces that provide the framework of the Laurentian supercontinent (Hoffman, 1988, 1989). The 'Rae-Hearne' Cratons are flanked to the west by the ca. 2.0–1.93 Ga Thelon-Taltson Magmatic Zone (Hoffman, 1988; Chacko et al., 2000) and to the southeast by ca. 1.95–1.8 Ga rocks comprising the Trans-Hudson Orogen (Hoffman, 1988; Lewry and Collerson, 1990). These two orogenic events resulted in pronounced Paleoproterozoic tectonic reworking of intervening Archean rocks throughout the Churchill Province (Hoffman, 1988, 1989; Ashton et al., in press).

Numerous Paleoproterozoic siliciclastic basins overlie the Archean rocks in the Churchill Province and represent evidence of this crustal reworking (Fraser et al., 1970; Ashton et al., in press). Some such basinal successions include contemporaneous mafic igneous rocks, reflecting localized extensional magmatism. An important example, covering a large expanse of the east-central Rae Craton (Fig. 1), is the Baker Lake Basin and igneous rocks of the Christopher Island Formation (e.g. LeCheminant et al., 1987; Peterson et al., 2002). Additionally, the Nonacho Group sedimentary

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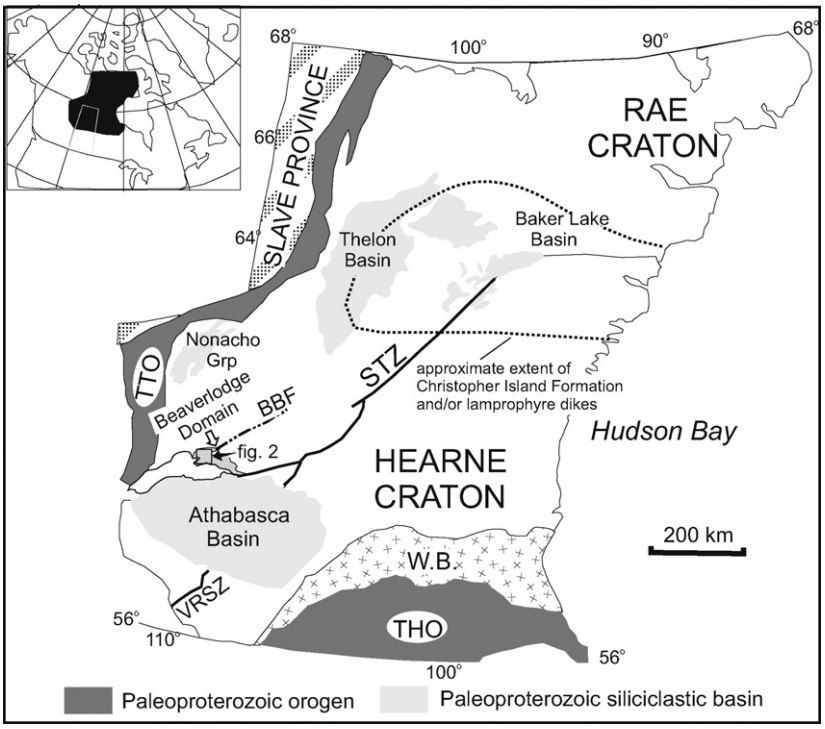


Fig. 1. Generalized geologic map of the Churchill Province and bounding orogens showing location of Proterozoic intracratonic basins in the Rae and Hearne Cratons (after Hoffman, 1988; TTO = Thelon-Taltson Orogen; THO = Trans-Hudson Orogen; WB = Wathaman Batholith; BBF = Black Bay Fault; STZ = Snowbird Tectonic Zone; VRSZ = Virgin River Shear Zone).

rocks of the western Rae Craton are spatially proximal to the Sparrow mafic dike swarm (McGlynn et al., 1974), though the exact genetic relationship, if any, is currently unknown. Though slightly younger, the Thelon and Athabasca Basins represent Proterozoic sedimentation centers, the former of which is also associated with mafic (shoshonitic) magmas (Peterson, 1995).

The Paleoproterozoic Martin Group, situated in northwestern Saskatchewan and the focus of this study, represents a Paleoproterozoic extensional basin that predates sedimentary rocks of the Athabasca Group, which are known to have been deposited after 1740 Ma (Rainbird et al., 2007). The Martin Group (MG), located in the Beaverlodge Domain in the Uranium City area (Fig. 1), records a ca. 5000 m-thick, siliciclastic-dominated red bed sequence with subordinate intercalations of mafic volcanic flows and sills. It has been suggested that a locally extensive suite of east- to southeast-trending mafic dikes acted as feeders to the MG mafic flows and sills (Tremblay, 1972), due to striking lithological similarities and field relationships that suggest close temporal relationships. Previous studies of the MG provided lithological and structural descriptions (Tremblay, 1972), stratigraphic characteristics (Mazimhaka and Hendry, 1984), and suggested several modes of basin formation (Fraser et al., 1970; Ramaekers, 1981; Ashton et al., 2001). The objective of the present study is to elucidate the age and the geochemical and Sm–Nd isotopic characteristics of these three mafic igneous suites (i.e. volcanic rocks, sills, and dikes) in order to constrain their tectonic setting of emplacement and to assess possible intersuite genetic relationships and magma source characteristics. This information augments our understanding of the tectonic evolution of the Churchill Province in northern Canada and helps to characterize the nature of the underlying crust and lithospheric mantle.

2. Regional geology

The MG is discontinuously exposed over an area of ~3000 km^2 (Mazimhaka and Hendry, 1984), with most exposures comprising small, fault-controlled successions. The main remnant of the depositional basin, covering ~250–300 km^2, is located immediately east of Uranium City and is the only portion of the MG known to host coeval igneous rocks (Fig. 2). The basin was fault-controlled and deposited over a highly irregular unconformity surface that varies in character from a sharp contact to local regolith (Tremblay, 1972; Mazimhaka and Hendry, 1984). The Archean to Paleoproterozoic basement to the MG encompasses an array of metamorphosed rock types and is referred to herein as 'undifferentiated basement' (Fig. 2). Comprehensive descriptions of basement units are provided by Tremblay (1972), Hartlaub (1999), and Ashton et al. (2000, 2001), and a brief summary of the principal basement lithologies are provided below.

The oldest rocks in the region comprise the 'Basement Complex' (Persons, 1983; Hartlaub et al., 2004) which represents a number of granitoid inliers including the Lodge Bay (3060 ± 40 Ma; Hartlaub et al., 2004), Elliot Bay (3014 ± 10 Ma; Persons, 1983) and Cornwall Bay (2999 ± 7 Ma; Hartlaub et al., 2004) granites. Three suites of Neoarchean to Paleoproterozoic intrusive rocks have also been identified as basement to the MG. The oldest of these is a ca. 2.6 Ga suite of granitic to tonalitic plutons (Hartlaub et al., 2004). A second intrusive episode at ca. 2.3 Ga is represented by the granitic to gabbroic 'North Shore Plutons' (O'Hanley et al., 1994; Hartlaub, 2004), which might have originated during terrane accretion related to subduction on the western margin of the Rae Craton (Hartlaub et al., 2007). The Basement Complex is unconformably overlain by an extensive amphibolite facies supracrustal sequence, consisting of basal quartzites, carbonates, ultramafic to intermediate igneous rocks, iron formations, and clastic sedimentary rocks, collectively referred to as the Murmac Bay Group (Macdonald, 1983). The Murmac Bay Group is of uncertain age and origin, but has been hypothesized as a rift succession (Wilson, 1998; Hartlaub et al., 2004) that was deposited after 2330 Ma (Ashton and Hunter, 2003). The final period of igneous activity prior to deposition of the MG is represented by a ca. 1930 Ma suite of crustally derived, pink to

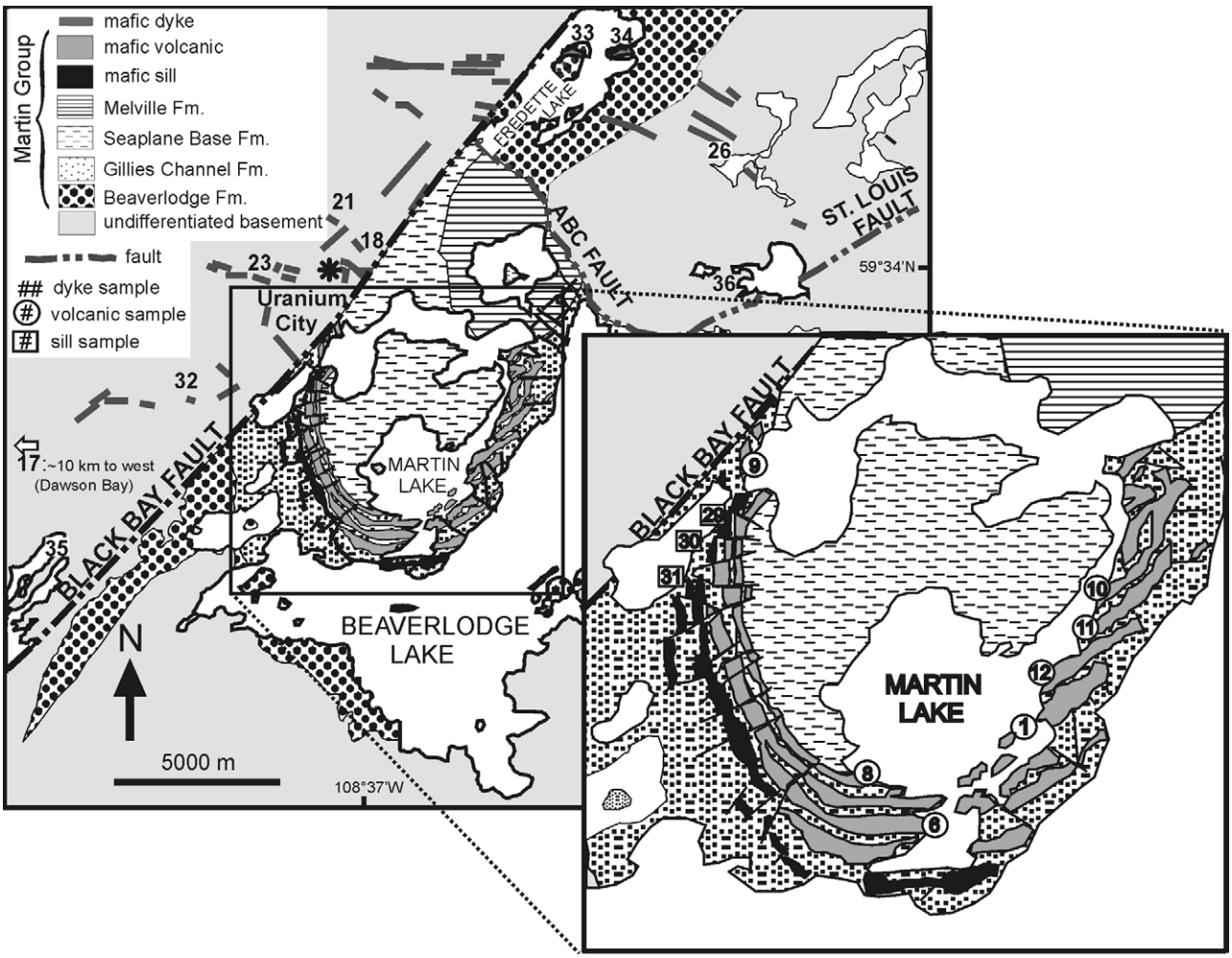


Fig. 2. Geology of the Martin Basin (after Mazimhaka and Hendry, 1984), showing location of samples collected for analysis in this study.

white leucogranites (Hartlaub et al., 2005), which also intrude and provide a minimum age bracket for the Murmac Bay Group.

Rocks of the basement record several episodes of deformation and metamorphism that occurred prior to Martin Basin formation and are manifested as variably oriented penetrative fabrics and fold generations. Basement rocks were subject to at least two and as many as three distinct metamorphic events, each of which attained greenschist to granulite facies P-T conditions (Ashton et al., 2001; Hartlaub, 2004).

3. Geology of the Martin Basin and associated igneous rocks

The main remnant of the Martin Basin (Fig. 2) is bound and truncated on its western margin by the Black Bay Fault, a prominent zone of brittle–ductile deformation that has a history of repeated reactivation (Krupicka and Sassano, 1972; Bergeron, 2001). The basin itself is hypothesized to represent a pull-apart structure (Ramaekers, 1981; Kraus and Ashton, 2000; Ashton et al., 2001) that formed in response to transtensional faulting in the upper crust. This faulting originated during simultaneous orogenesis on opposing continental margins, thus imposing a vise-like stress regime on the continental interior (Ashton et al., in press). Within the basin, the MG displays a generally fining-upward stratigraphic sequence of redbeds consisting of basal conglomerates through to coarse sandstone, fine sands, and finally to siltstone at the top of the sequence. Fig. 3 shows the subdivision of the MG into four formations (Langford, 1981; Elliott, 1982): the Beaverlodge, Gillies Channel, Seaplane Base, and Melville Lake Formations, in order of decreasing depositional age. The sedimentological character of the MG is fully described by Mazimhaka and Hendry (1984); we therefore limit the following description to the associated intrusive and extrusive igneous rocks.

3.1. Mafic volcanic rocks

Plagioclase-phyric mafic volcanic rocks comprise a series of at least six ~125 m thick flows, each recording two or more volcanic eruptions (Tremblay, 1972). The flows are interbedded with arkose and conglomerate of the Gillies Channel Formation, indicating brief periods of subsidence and sedimentation between eruptions. Although volcanism is believed to have occurred predominately in a subaerial environment (Tremblay, 1972), rare structures that have a pillow-like morphology are also noted (Fig. 4a). Flow-top breccias (Fig. 4b) and increased concentrations of vesicles and amygdules at the top of individual flows provide local way-up indicators. The amygdules (typically 0.5–10 mm diameter) have infillings of calcite, chlorite, cryptocrystalline quartz, or a combination thereof (e.g. Fig. 4c).

The mineralogical composition of the mafic volcanic rocks is consistent regardless of stratigraphic position. Plagioclase is abundant, comprising ~50–60% of the total mineral content. Euhedral feldspar phenocrysts or "laths" reaching lengths of up to 0.5 mm are common (Fig. 4d), though several samples instead exhibit a homogranular texture. Plagioclase composition is reported as

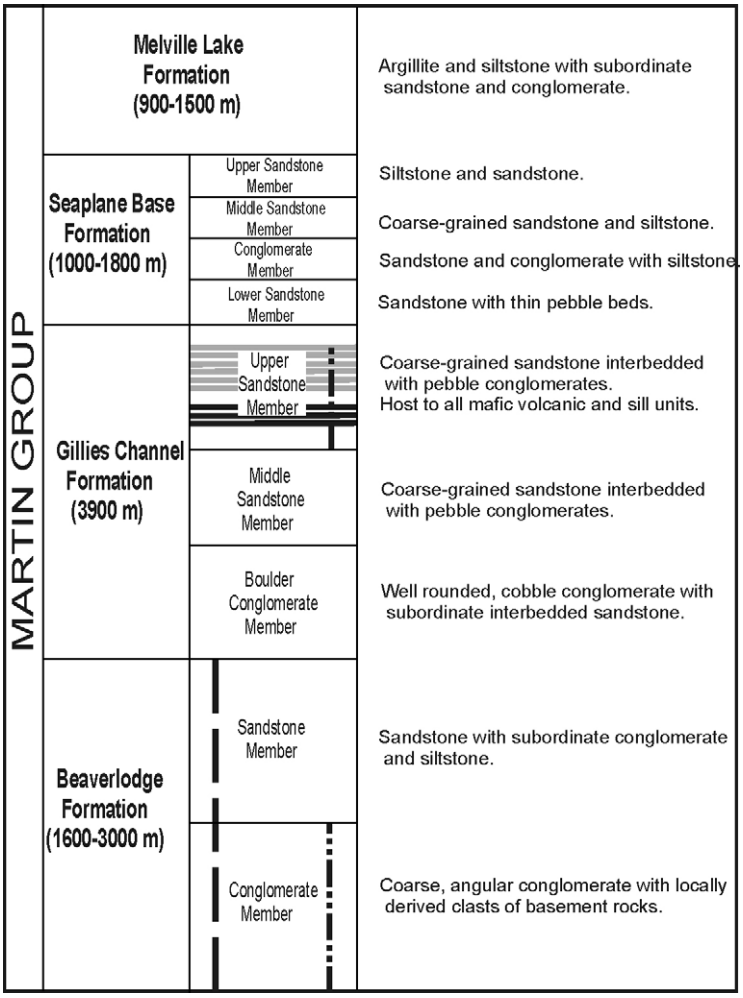


Fig. 3. Stratigraphic section for the Martin Group (following Langford, 1981; Elliott, 1982) with corresponding lithologic descriptions (Mazimhaka and Hendry, 1984). Dashed vertical lines and dashed + dotted vertical lines represent known stratigraphic extent of MG mafic dikes, and vein-hosted uranium mineralization, respectively. In the Upper Sandstone Member of the Gillies Channel Formation, black and grey horizontal lines show the stratigraphic position of the Martin Group sills and volcanic flows, respectively.

An_{35} (Tremblay, 1972). Sanidine, though rare, was also noted in some samples. Clinopyroxene, like feldspar, is also present both as phenocrysts up to 0.5 mm in length and within the very fine to fine-grained matrix, and constitutes ~25% of the volcanic rocks. Although unaltered clinopyroxene is common, the majority of crystals have been partially or completely chloritized (Fig. 4e). Minor minerals include 5–10% magnetite and ilmenite, and 1–3% apatite. The remainder of the rock, comprising ~10% of the mode, consists of interstitial brown to green material that is likely devitrified glass.

3.2. Mafic sills

At least three distinct sills are exposed within the MG, but it is possible that more exist to the south beneath Beaverlodge Lake. The MG sills are observed to intrude only sedimentary rocks of the Gillies Channel Formation. They are nearly indistinguishable from the mafic volcanic rocks, except for the noticeably larger size of feldspar phenocrysts (Fig. 4f), typically 5–10 mm in length, and by their slightly discordant relationship with sedimentary interbeds of the Gillies Channel Formation. Similarly, the mineralogical makeup of the sills is nearly identical to that of the volcanic rocks, contrasting only in their increased grain size (typically medium to coarse-grained; Fig. 4f–h). Rare angular blocks (<15 cm) of arkose and conglomerate are found within the sills near the contact with the host units.

The characteristic mineral composition of the sills is approximately 55–65% feldspar, 15–20% clinopyroxene or chloritized relicts thereof, 5–10% Fe–Ti oxides, and <5% combined apatite and aphanitic (glassy) groundmass. All examined samples were porphyritic, though phenocryst abundances ranged from ~5 to 30%. Feldspar and clinopyroxene are present as randomly oriented sub- to euhedral phenocrysts as well as in the fine-grained groundmass. Feldspar composition is predominately andesine-labradorite ($An_{35–55}$; Christie, 1952). Sanidine is also noted both as phenocrysts and in the groundmass in some samples. Euhedral apatite crystals are completely enclosed within clinopyroxene phenocrysts (Fig. 4h). Amygdules of calcite and chlorite are rarely present.

3.3. Mafic dikes

Mafic dikes are present over an extensive area of the western Beaverlodge Domain (Fig. 2) and the Zemlak Domain to the west, and crosscut all rock types of the undifferentiated basement (e.g. Fig. 4i). These dikes, designated as the 'Uranium City dikes' (Ashton et al., in press) are steeply dipping to subvertical, are most commonly east-west trending and were emplaced along pre-existing, conjugate fracture sets (Tremblay, 1972; Ashton et al., 2001). The dikes typically range from 1 to 50 m in width and are <50 m long, but one dike with a length of 1000 m is reported (Tremblay, 1972). Their size and shape varies along strike, commonly pinching and swelling.

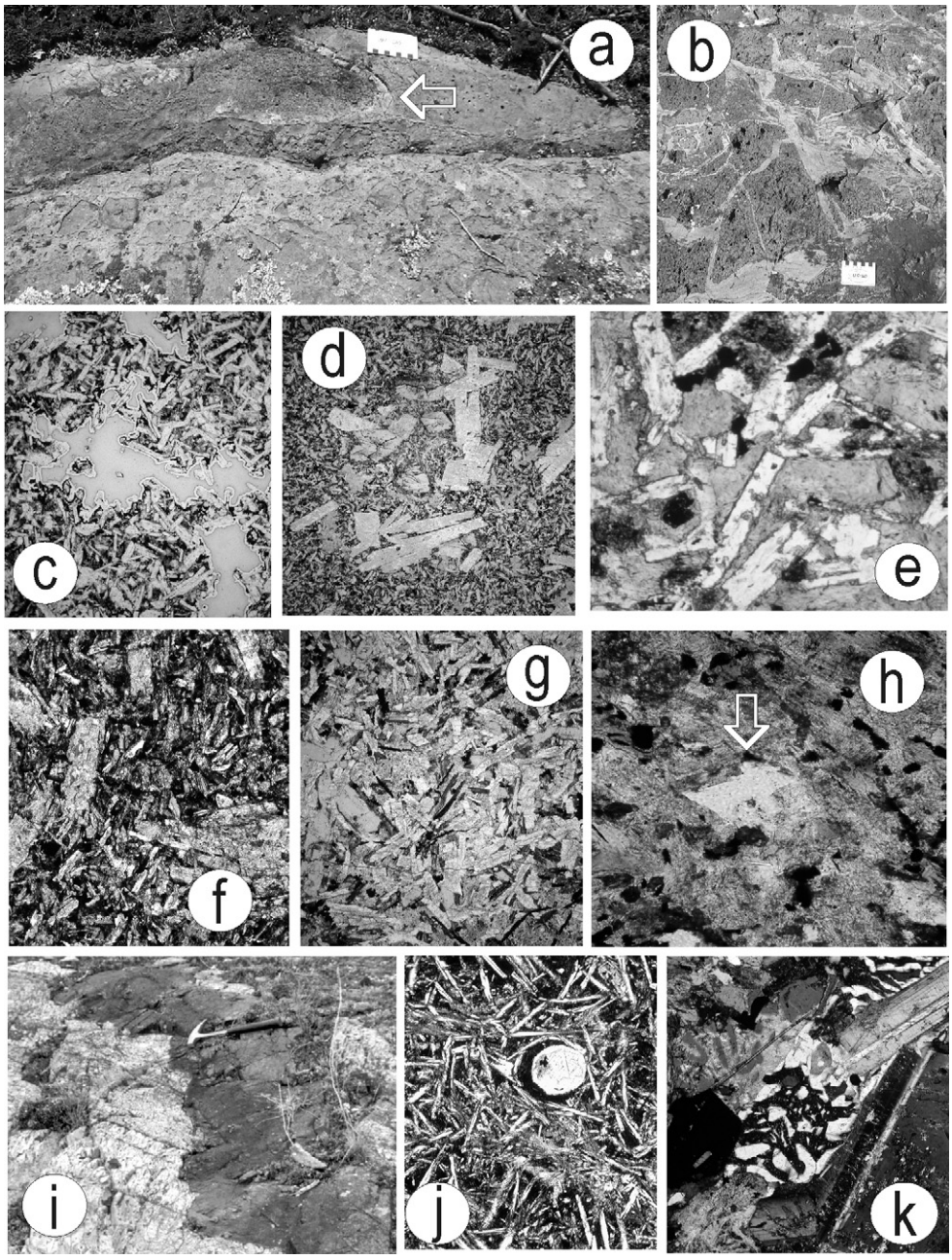


Fig. 4. Photographs and photomicrographs of MG igneous rocks: (a) possible pillow in volcanic flow; (b) volcanic flow-top breccia; (c) chlorite-filled amygdule rimmed with cryptocrystalline quartz in volcanic rock (FOV = 5.5 mm); (d) plagioclase feldspar phenocrysts in volcanic rock (field of view (FOV) = 11 mm); (e) apatite enclosed within chloritized clinopyroxene in sill (FOV = 7 mm); (f) medium grained groundmass surrounding feldspar phenocrysts in sill (FOV = 2 mm); (g) typical mineralogy and texture of sills (FOV = 2 mm); (h) primary quartz (FOV = 1 mm) in mafic dikes; (i) mafic dike cutting felsic phase of 'undifferentiated basement; (j) quartz-filled amygdule surrounded by plagioclase feldspar laths in mafic dike (FOV = 5.5 mm); and (k) myrmekitic texture in dike (FOV = 15 mm).

Contacts with the wall rock are sharp, and the dikes rarely contain small (<5 cm) angular wall rock xenoliths in their margins. The mafic dikes cut fabrics resulting from at least three earlier phases of deformation (Ashton et al., 2001) but, as with the MG, have been folded during a subsequent deformational event.

Despite the abundance of dikes in the region, the only two known to intrude the MG crosscut the basal Beaverlodge Formation at Fredette Lake (Fig. 2). One small dike was also reported to cut one of the MG sills (Christie, 1952), although this relationship was not confirmed in the present study due to an inability to locate this dike.

The dikes vary widely in grain size from aphanitic to coarse, and locally exhibit amygdaloidal (Fig. 4j), porphyritic (Fig. 4k) and/or diabasic (Fig. 4j) textures. In spite of this textural variability, the mineral compositions of the dikes are similar to each other, as well as to the volcanic rocks and sills. Primary mineralogical constituents include plagioclase (50–60%), clinopyroxene (20–30%), Fe–Ti oxides (5–10%) quartz (<1%) and apatite (<1%). Myrmekitic patches were also noted in one sample (RM-17d; Fig. 4k).

3.4. Structure and metamorphism/alteration

The Martin Group has been folded into a prominent north-trending synclinal structure, most noticeable from the map pattern of the volcanic rocks and sills (Fig. 2). Ashton et al. (2001) interpret this fold as a late (F4) structure related to east–west shortening of

the region and invoke a complex model linking basin development to the early stages of this (D4) compressional event. The same fold generation also affected the Uranium City dikes. Subsequent to this folding, post-D4, east-west extension down-dropped the area east of the Black Bay Fault and south of the ABC Fault, thereby preserving the currently exposed remnant of the basin. Additionally, a series of late, variably oriented brittle normal faults cut the MG that disrupt the lateral continuity of the volcanic and sill units (Fig. 2).

The most significant metamorphic events in the region occurred prior to deposition of the MG (Ashton et al., 2001; Hartlaub, 2004; Ashton et al., in press). However, sericite along the bedding planes of MG sedimentary rocks (Tremblay, 1972) provides evidence for a post-depositional episode of sub-greenschist facies metamorphism. Additionally, extensive hydrothermal alteration, probably representing multiple fluid flow events, affects all three igneous suites. Vesicles are commonly filled with a variety of hydrothermal minerals including chlorite, cryptocrystalline quartz, and calcite and there is extensive sericitization/illitization of feldspar, chloritization of clinopyroxene, and hematitization. The intensity of alteration is highly variable, ranging from negligible to complete replacement of primary minerals. This hydrothermal activity is probably also linked to the generation of several vein-hosted uranium deposits in the region (Mazimhaka and Hendry, 1989; see Fig. 3).

4. Sampling and analytical protocol

4.1. Whole rock geochemistry

An attempt was made to obtain samples from each volcanic unit and sill in the stratigraphic section, though this was difficult due to the lateral displacement of units by late normal faults (Fig. 2). In all, 33 samples were collected for geochemical analysis, including 14 volcanic, 3 sill, and 16 dike samples. In an attempt to avoid highly altered rocks, the majority of samples were obtained using a rock saw. Thirteen of the samples were rejected after the presence of pervasive alteration was noted during petrographic analysis. The remaining 20, including 7 volcanic, 3 sill, and 10 dike samples (see Fig. 2 for sample locations) were used for whole rock geochemical analysis.

The samples were crushed and ground in agate and the major element oxides and selected trace elements were determined by X-ray fluorescence spectrometry at X-Ray Assay Laboratories (Don Mills, Canada). The same rock powders were analyzed for their REE and HFSE content using inductively coupled plasma mass spectrometry (PerkinElmer 5000) at the Department of Geological Sciences at the University of Saskatchewan. Powder aliquots of 200 mg were dissolved using the Na_2O_2 sinter technique (Longerich et al., 1990) to reduce the possibility that refractory minerals would be incompletely dissolved. Analysis of reagents and procedural blanks yielded concentration levels of <10 ppb for REEs, Hf, Zr and Nb. Detection limits, defined as 3 sigma of the procedural blank, were less than 1 ppm for all trace elements.

4.2. Geochronology/Isotopic tracing

A 1 kg sample from a ca. 50 cm wide mafic dike (RM-37d) was collected from 'Fold Island', located ~19 km southwest of Uranium City (UTM: 617278E, 6598846N), for U–Pb geochronology. The sample was crushed and ground and the heavy minerals were separated using a Wilfley water table. The heavy mineral separate was sieved to obtain uniformly fine material, and highly magnetic particles were removed using a high-strength magnetic field. Final splitting of the sample was carried out by heavy liquid (methyl iodide) and Frantz magnetic separation techniques. Individual fractions of baddeleyite were handpicked from the separates with the aid of a binocular microscope. Uranium and lead were isolated using the anion exchange chromatography method developed by Krogh (1973). Estimated analytical blanks for the procedure were 2 pg for Pb and 0.5 pg for U. The U and Pb were loaded together onto a Re filament following the Si-gel technique of Cameron et al. (1969). All isotopic ratios were measured using a Daly photomultiplier detector in single-collector mode on a VG354 mass spectrometer. All uncertainties reported in Table 2 are 1σ and were calculated using an in-house program that numerically propagates all known sources of uncertainty. Regression calculations and age uncertainties were determined using Isoplot version 3.0 (Ludwig, 2003).

For Sm–Nd isotopic analysis, an aliquot of the powdered samples used for major and trace element analyses were spiked with an appropriate amount of a mixed ^{149}Sm–^{150}Nd tracer solution prior to digestion in doubly distilled HF-HNO_3 in closed screwtop Savillex beakers. After digestion, the solution was dried down and then redissolved in doubly distilled 6.1N HCl to convert the salts to chlorides. Both Sm and Nd were separated using conventional cation exchange columns employing a two-stage procedure, prior to loading with phosphoric acid on Re double filaments. Isotopic analysis of Sm and Nd were performed using a Finnigan MAT 261 multi-collector mass spectrometer in static mode. Spike unmixing and mass fractionation corrections were performed offline and used the following normalization ratios: $^{148}Sm/^{154}Sm = 0.49419$ and $^{146}Nd/^{144}Nd = 0.7219$. During the course of this project, analyses of the La Jolla Nd standard yielded a $^{143}Nd/^{144}Nd$ value of 0.511834 ± 18 (2σ, $n = 10$), and procedural blanks were approximately 100 pg Nd and 30 pg Sm. The following reference values were used to calculate initial epsilon Nd and depleted mantle values: $^{143}Nd/^{144}Nd_{CHUR\ (present)} = 0.512638$; $^{147}Sm/^{144}Nd_{CHUR\ (present)} = 0.1966$; $^{143}Nd/^{144}Nd_{depleted\ mantle\ (present)} = 0.51316$; and $^{147}Sm/^{144}Nd_{depleted\ mantle(present)} = 0.2141$ (Goldstein et al., 1984).

5. Results

5.1. Whole rock geochemistry

(i). Element mobility

Major and trace element compositions of the 20 samples are listed in Table 1. Due to the variably altered nature of the igneous rocks (henceforth collectively referred to as the 'Martin Group Igneous Rocks' (MGIR) for simplification, with no genetic connotations between suites implied), samples were screened for evidence of chemical mobilization. This was accomplished by plotting of major and trace element abundances for all samples against elements with moderate ionic potentials (e.g. Zr), which are typically immobile in hydrothermal solutions (Cann, 1970). Plots of abundances of CaO, Na_2O, and MgO vs. Zr for all analyzed samples display a significant degree of scatter (Fig. 5a–c), indicating that elements of low ionic potential, including large ion lithophile elements (LILE), were mobilized by fluid movement, thus compromising their validity for geochemical discrimination. In contrast, plots of elements of moderate ionic potential, including Nb, Th, and Ce vs. Zr (Fig. 5d–f), display significantly greater correlation, indicating that these elements remained relatively immobile during secondary alteration processes. One dike (RM-18d) sample deviates from the obvious trends exhibited by these plots (labeled in Fig. 5) suggesting possible disturbance of the primary chemical composition of this rock by hydrothermal fluids. Consequently, this sample was not used for geochemical characterization.

Table 1
Representative major and trace element analyses of the MGIR. Last character of sample name denotes analysis of volcanic (v), sill (s), or dike (d).

	RM-1v	RM-6v	RM-8v	RM-9v	RM-10v	RM-11v	RM-12v	RM-29s	RM-30s	RM-31s	RM-17d	RM-18d	RM-21d	RM-23d	RM-26d	RM-32d	RM-33d	RM-34d	RM-35d	RM-36d
SiO_2 (%)	50.8	52.5	43.3	50.3	46.7	50.0	52.1	52.5	55.3	52.4	52.7	50.5	51.7	52.1	52.6	50.5	52.9	52.4	51.2	53.3
TiO_2	1.5	1.4	1.8	2.5	2.9	1.8	1.8	3.1	2.3	2.1	1.9	1.7	1.7	1.2	1.3	1.9	1.3	1.3	1.9	1.3
Al_2O_3	14.4	15.2	15.6	14.9	16.0	15.9	15.4	13.7	14.5	14.2	15.0	14.3	14.5	14.6	14.9	14.2	14.8	15.2	14.5	14.4
Fe_2O_3	12.2	11.0	10.8	10.7	12.8	10.8	12.0	10.7	9.8	11.1	11.5	10.9	12.0	12.0	11.5	11.5	11.9	11.2	13.4	11.8
MnO	0.17	0.20	0.21	0.09	0.11	0.14	0.15	0.08	0.16	0.29	0.08	0.14	0.16	0.18	0.33	0.08	0.21	0.20	0.06	0.16
MgO	6.3	6.3	7.3	3.7	5.6	6.2	4.8	5.2	4.0	5.2	6.2	7.3	5.9	5.8	5.2	5.4	5.5	5.6	9.0	5.1
CaO	4.3	4.8	7.0	4.3	4.4	3.9	3.1	3.3	5.8	7.0	4.8	4.3	6.3	6.8	7.9	7.7	7.9	5.6	1.3	8.6
Na_2O	4.9	4.7	4.7	2.7	5.1	4.2	3.7	2.5	2.4	2.4	2.2	3.1	2.3	2.2	2.2	2.3	2.2	2.7	0.37	2.3
K_2O	0.35	1.3	0.62	4.3	1.6	1.8	3.0	3.0	2.3	1.9	2.1	1.9	2.0	2.1	1.6	1.3	1.7	1.5	1.6	1.2
P_2O_5	0.50	0.47	0.85	1.5	1.5	0.86	0.86	1.7	1.0	0.89	0.76	1.0	0.72	0.40	0.35	0.60	0.33	0.35	0.60	0.34
L.O.I.	4.8	2.3	7.8	4.0	3.0	4.5	2.9	3.5	2.1	2.2	2.5	4.8	2.7	2.0	2.2	4.5	1.2	3.9	6.1	1.8
Total	100	100	99.9	99.0	99.6	100	99.8	99.3	99.6	99.7	99.7	99.8	99.8	99.5	100	99.9	99.8	99.9	100	100
Rb (ppm)	0.86	1.44	2.89	2.54	17.2	4.55	13.8	4.11	9.23	6.01	6.54	10.5	11.2	14.2	5.50	2.92	4.58	4.13	6.89	3.64
Ba	310	566	560	3450	2500	830	1830	2740	2550	2878	2022	2136	2411	2135	1146	1675	1446	1776	90.4	916.4
Sr	234	709	340	393	647	353	553	319	595	657	671	297	762	693	460	641	474	377	53.9	429
La	68.4	77.2	99.3	205	218	103	105	237	126	112	108	198	93.2	70.2	54.5	93.2	53.6	56.3	71.5	56.0
Ce	143	98.6	219	426	452	224	219	487	264	235	220	382	195	136	106	186	105	111	151	109
Pr	16.4	17.2	25.0	48.1	51.3	26.0	25.3	54.0	30.4	27.5	24.8	41.6	22.8	15.1	11.9	21.4	11.9	12.5	17.3	12.3
Nd	60.9	63.5	95.7	174	187	99.0	94.3	196	113	104	90.1	149	86.3	56.4	43.9	79.3	44.6	46.5	65.3	45.2
Sm	9.56	10.1	14.4	24.7	27.6	15.2	14.4	29.1	17.4	15.8	14.3	20.6	13.3	8.79	7.33	12.8	7.25	7.70	11.3	7.71
Eu	2.31	2.29	3.32	4.97	5.80	3.32	3.13	5.25	4.10	3.60	3.37	4.57	3.08	2.14	1.79	2.93	1.91	1.97	2.27	1.96
Gd	7.22	7.48	9.54	14.8	17.2	9.98	9.42	18.1	11.3	10.5	9.64	12.1	8.87	6.32	5.79	9.15	5.79	6.12	8.46	5.98
Tb	0.83	0.85	0.97	1.36	1.64	1.01	0.99	1.69	1.15	1.08	1.03	1.07	0.92	0.71	0.69	1.00	0.70	0.72	1.03	0.73
Dy	4.47	4.7	4.99	6.61	7.82	5.35	5.04	8.03	5.82	5.55	5.33	5.05	4.75	4.01	3.95	5.38	4.01	4.16	5.75	4.14
Ho	0.82	0.85	0.88	1.09	1.26	0.94	0.89	1.28	0.97	0.94	0.9	0.8	0.83	0.72	0.70	0.94	0.75	0.74	1.09	0.76
Er	2.23	2.23	2.26	2.68	3.08	2.39	2.29	3.06	2.38	2.34	2.32	1.99	2.09	1.97	1.89	2.4	1.99	2.06	2.97	2.06
Tm	0.31	0.30	0.31	0.35	0.39	0.34	0.3	0.39	0.31	0.3	0.3	0.25	0.29	0.27	0.26	0.32	0.28	0.27	0.43	0.29
Yb	1.89	1.87	1.84	2.1	2.25	2.00	1.81	2.23	1.83	1.82	1.87	1.48	1.65	1.59	1.60	1.98	1.67	1.68	2.67	1.72
Lu	0.36	0.36	0.35	0.38	0.42	0.38	0.34	0.42	0.34	0.35	0.37	0.28	0.3	0.32	0.31	0.39	0.34	0.33	0.52	0.34
Y	19.8	23.3	22.3	26.0	32.5	23.8	22.8	34.9	23.2	24.9	23.8	20.7	21.1	19.3	18.3	25.2	19.4	20.1	28.1	19.6
Zr	190	189	286	468	556	297	299	778	330	289	233	357	203	175	142	217	144	156	210	150
Th	4.72	3.31	7.94	11.5	12.7	8.19	7.67	15.81	7.38	5.94	5.56	20.6	5.18	5.85	4.67	4.58	4.54	4.90	4.84	5.23
Hf	5.49	5.25	7.54	13.2	14.4	8.05	8.08	17.9	9.63	7.91	6	9.88	5.61	4.91	4.24	5.87	4.45	4.8	6.42	4.85
Nb	12.0	7.52	18.8	35.5	31.0	18.9	19.6	35.0	20.0	18.2	12.0	19.8	13.2	9.34	7.62	10.6	7.73	8.29	9.62	8.3
Ta	0.78	0.54	1.08	2.17	1.87	1.18	1.16	1.84	1.2	0.98	0.66	1.19	0.72	0.52	0.44	0.54	0.45	0.47	0.58	0.50
U	0.36	0.46	0.58	1.70	0.93	0.55	0.72	1.00	4.41	0.69	0.48	2.05	0.59	0.55	0.42	0.47	0.47	0.5	1.37	0.63

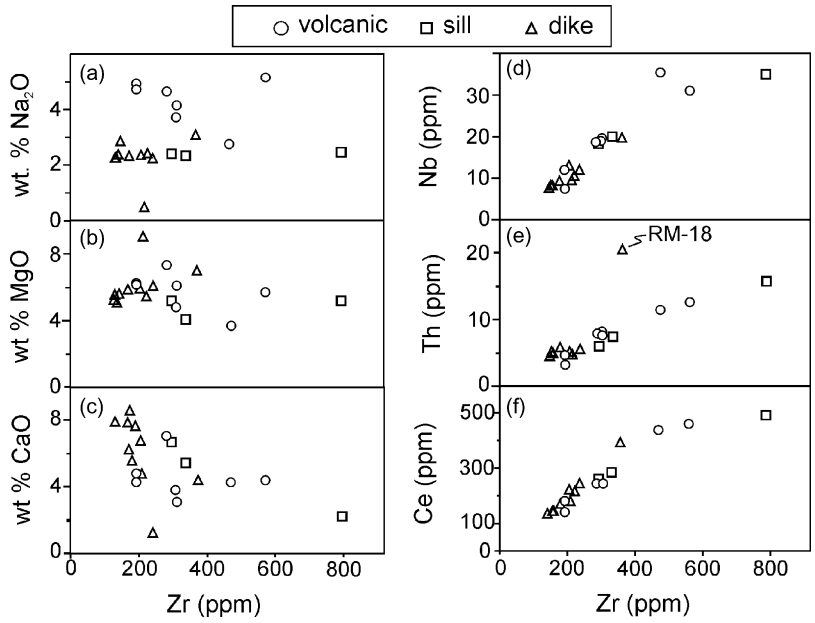


Fig. 5. Binary plots of Zr vs. (a) Na_2O, (b) MgO, (c) CaO, (d) Nb, (e) Th, and (f) Ce for the Martin Group igneous rocks, utilized to detect possible effects of alteration on primary chemical compositions (following Cann, 1970). Noticeable scatter of samples in (a), (b), and (c) might reflect mobilization of these large ion lithophile elements, whereas clustering and/or linear trending of samples in (d), (e), and (f) is consistent with preservation of primary compositions for these elements characterized by intermediate ionic potentials.

(ii). Rock classification

Although the geochemical discrimination of the MGIR presented here focuses primarily on trace element compositions, the total alkali vs. silica binary plot (LeBas et al., 1992; not shown) can provide a general compositional characterization of the rocks. Using this plot, the Uranium City dike samples are characterized as basalts to basaltic andesites, the sills as basaltic andesites to basaltic trachyandesites, and the volcanic rocks as tephrites to trachybasalts to basaltic trachyandesites. Based upon their relative Na_2O and K_2O contents, the dikes can be further subclassified as subalkaline, the sills as potassic trachybasalts to shoshonites (high K), and the volcanic rocks as predominantly hawaiites or mugearites (high Na; LeBas et al., 1992). Notwithstanding the possibility that some of these classifications do not accurately reflect primary chemical compositions, noteworthy are the observations that: (1) the dikes and volcanic rocks tend to cluster as individual groups in distinct compositional regions, with the sills being transitional between these two suites, and (2) despite an overall similarity in composition, there is a progression towards an increasingly alkaline character from the dikes to the sills and finally to the volcanic rocks.

A plot of the MGIR on a rock type classification diagram utilizing immobile trace elements (Zr/TiO_2 vs. Nb/Y; Winchester and Floyd, 1977) yields results generally consistent with the estimated modal mineralogy and major element classifications. This diagram (Fig. 6) reveals an overall compositional similarity between the three suites, though subtle differences are discernible. As with the major element plot, the dikes tend to cluster on the border between the subalkaline basalt and andesite fields, whereas relatively more alkaline and more evolved compositions generally characterize the sills and volcanic rocks.

(iii). Rare earth element/trace element patterns

Chondrite-normalized rare earth element (REE) patterns for samples of the three suites are shown in Fig. 7a. The rocks are all characterized by a substantial enrichment of all REEs relative to chondrite and an overall negative slope due to pronounced enrichment of the light rare earth elements (LREE) relative to the heavy rare earth elements (HREE), with $(La/Yb)_N$ ratios ranging from 22 to 72. The HREE segments of the patterns are all relatively flat, with $(Gd/Yb)_N$ ratios ranging from 3 to 7. Although the patterns for the three suites are nearly identical in shape, the dikes are consistently the least LREE enriched of the three suites (La/Yb_N = 22–39), compared to the volcanic rocks (24–66) and sills (42–72).

Similar enrichment patterns are likewise reflected in primitive mantle-normalized extended trace element patterns for the three suites (Fig. 7b). As with the REE concentrations of these rocks, noticeable is a significant overall enrichment of all trace elements with respect to primitive mantle. The most distinctive feature of the patterns is the combined enrichment of incompatible elements (Th, LREE) and relative depletions in the high field strength elements (HFSE: Ta, Nb, Zr, and Ti), manifested as pronounced negative anomalies in the overall pattern. Fractionation of the incompatible elements from elements retained in the melt residue is indicated by the progressively decreasing concentrations of elements towards the right side of the diagram.

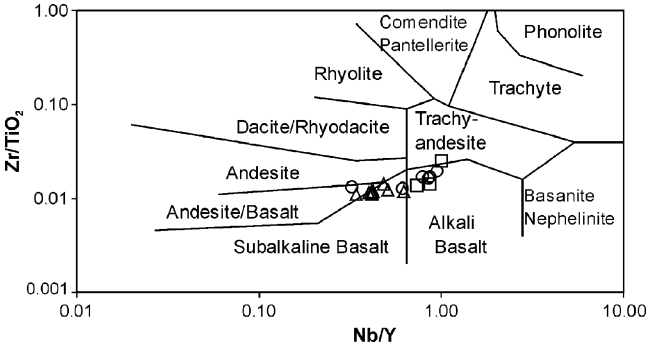


Fig. 6. Rock type classification diagram utilizing trace element contents (Zr/TiO_2 vs. Nb/Y; Winchester and Floyd, 1977).

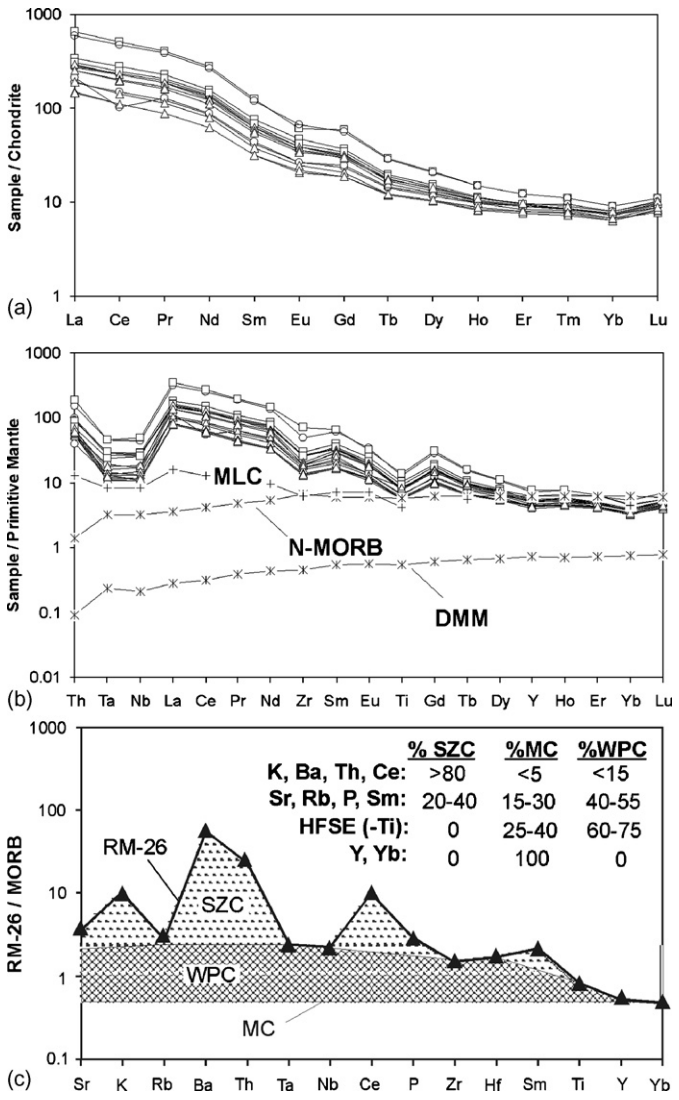


Fig. 7. (a) Chondrite – (Sun and McDonough, 1989) normalized REE patterns of volcanic rocks (circles), sills (squares), and dikes (triangles); (b) Primitive mantle – (Sun and McDonough, 1989) normalized trace element patterns of volcanic rocks, sills, and dikes. Also shown for comparison is the composition of depleted MORB mantle (DMM, Workman and Hart, 2005), normal MORB (N-MORB, McDonough and Sun, 1989) produced from melting of DMM, and mafic lower crust (MLC, Taylor and McLennan, 1985); (c) MORB – (Pearce et al., 1981) normalized trace element pattern of dike RM-26, representative of all Martin Group igneous rocks and dikes, and showing proportions of enrichment components (after procedure of Pearce, 1983).

5.2. Geochronology/isotopic tracing

(i). U–Pb Geochronology

All five analyzed fractions from mafic dike RM-37d (Table 2) consisted of 12–14 prismatic to platy baddeleyite crystals. Analyses 1 and 2 were clear grains, whereas analyses 3, 4 and 5 were slightly cloudy and had surface imperfections. The clarity of the grains is likely related to radiation damage, since analyses 3–5 have higher U concentrations (1352–1729 ppm) than analyses 1 and 2 (891–920 ppm). Analyses from the 5 fractions yield an upper intercept age of 1818 ±4 Ma (MSWD = 3.7; Fig. 8), which we interpret as the crystallization age of the dike. This result is generally consistent with an existing, though relatively imprecise, K–Ar biotite age of 1835 ± 50 Ma for a 'dark grey gabbroic dyke' from the Uranium City area (Wanless et al., 1967), presumed to be part of the Uranium City dike suite.

Table 2
U–Pb isotopic results for baddeleyite from dike sampleRM-37d.

Sample description[a]		Concentrations (ppm)					Isotopic ratios[d]				Apparent ages (Ma)				PD
		Weight (mg)	U	Th[b]	Pb rad.	Th/U ratio	Pb (pg)[c]	^{206}Pb ^{204}Pb	^{206}Pb ^{238}U	^{207}Pb ^{235}U	^{207}Pb ^{206}Pb	^{206}Pb ^{238}U	^{207}Pb ^{235}U	^{207}Pb ^{206}Pb	
1	B(12)	0.005	920	23	288	0.03	43	2202	0.3256 ± 7	4.994 ± 14	0.1112 ± 2	1817.1 ± 3.3	1818.4 ± 2.3	1819.8 ± 3.6	−0.39	
2	B(14)	0.005	891	39	278	0.04	123	685	0.3224 ± 6	4.936 ± 10	0.1110 ± 2	1801.6 ± 2.7		1808.5 ± 1.7	1816.4 ± 2.4	0.93
3	B(12)	0.002	1388	37	436	0.03	6	9638	0.3264 ± 4	4.992 ± 5	0.1109 ± 1	1820.8 ± 1.7		1817.9 ± 0.9	1814.6 ± 0.9	−0.39
4	B(12)	0.002	1352	35	423	0.03	5	8258	0.3254 ± 4	4.984 ± 5	0.1111 ± 1	1816.1 ± 1.8		1816.7 ± 0.9	1817.3 ± 1.0	0.08
5	B(13)	0.003	1729	50	545	0.03	12	8100	0.3276 ± 3	5.024 ± 5	0.1112 ± 0	1826.6 ± 1.6		1823.4 ± 0.9	1819.7 ± 0.6	−0.44

rad = radiogenic.
[a] *Abbreviations*: B, baddeleyite. The number in parentheses identifies the number of grains analyzed.
[b] Model Th concentration calculated from 208Pb/206Pb ratio and 207Pb/206Pb age.
[c] Total common lead in analysis.
[d] Atomic ratios corrected for 2.0 pg Pb and 0.5 pg U in blanks, fractionation, and initial common Pb (Stacey and Kramers, 1975). Uncertainties are reported at 1σ.

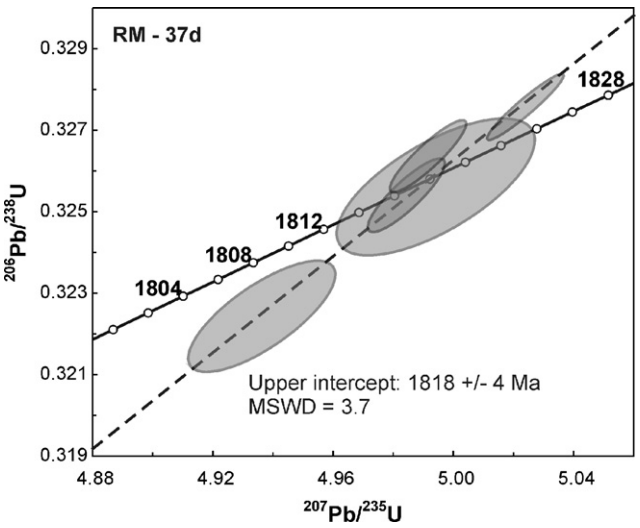


Fig. 8. Concordia plot of U–Pb baddeleyite data from dike RM-37d.

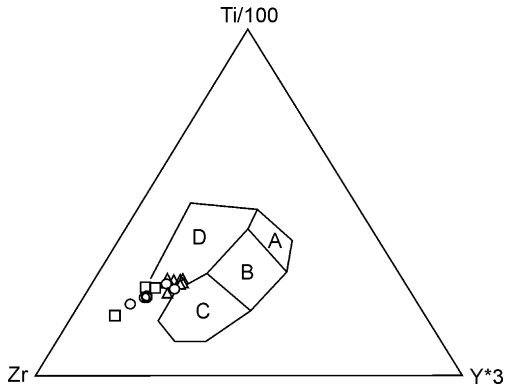


Fig. 9. Ternary plot of Zr vs. Ti/100 vs. Y*3 (Pearce and Cann, 1973) to elucidate tectonic setting of Martin Group igneous rocks. Fields are: (A) island arc tholeiite, (B) MORB, calc-alkali basalt, island-arc tholeiite, (C) calc-alkali basalt and (D) within-plate basalt.

(ii). Sm–Nd Isotopes

The results of Sm–Nd isotopic analyses for eight samples of the MGIR (3 volcanic, 2 sill, and 3 dike samples) are provided in Table 3. The analyzed samples represent the complete range of enrichment levels observed in the MG samples with respect to REE compositions. Initial $^{143}Nd/^{144}Nd$ values range from 0.510901 to 0.511104 for these samples, yielding a range of ε_{Nd} values from −4.7 to −6.3, assuming an age of 1818 Ma. These Nd isotopic compositions correspond to depleted mantle model ages (T_{DM}) ranging from 2427 to 2566 Ma (Table 3).

6. Discussion

6.1. Timing and tectonic setting of martin basin development

The U–Pb baddeleyite date and trace element data presented here substantiate the continental intraplate model for the origin of the basin, previously proposed due to the depositional style (e.g. Mazimhaka and Hendry, 1984) and structural history (e.g. Ashton et al., 2001) of the MG. As shown in Fig. 3, the only Uranium City dikes observed to within the MG crosscut the basal Beaverlodge Formation, stratigraphically below all known MG volcanic rocks. No dikes are observed to cut the Martin Group above the volcanic sequences, nor are any younger volcanic rocks known in the region. Together, these observations are consistent with the interpretation that all dike emplacement occurred prior to or during volcanism, but following deposition of the basal unit of the Martin Basin. The new 1818 Ma U–Pb baddeleyite age thus suggests that basin development, dike intrusion, and volcanism took place in an intracontinental setting, at which time rocks underlying the Beaverlodge Domain represented part of the Archean hinterland to the flanking Paleoproterozoic Thelon-Taltson and Trans-Hudson Orogens (Fig. 1).

This inference is further substantiated through use of a ternary plot of Zr vs. Ti/100 vs. Y*3 (Pearce and Cann, 1973), designed for discrimination of tectonic setting. When plotted on this diagram (Fig. 9), the MGIR are characterized by exceedingly high Zr concentrations relative to Ti and Y, placing them within or closest to the field characterizing 'within-plate basalt', and clearly outside of the 'calc-alkali basalt' field. This plot also reveals distinct subgroups within the MGIR in that, whereas the dikes cluster within the 'within-plate' field, the sills and volcanic rocks are pulled out of this field due to their higher Zr contents. This slight compositional difference, also evident in the trace element patterns, has implications for interpretation of the source and subsequent magmatic evolution of these three igneous suites, and will be discussed further in the following section.

6.2. Characterization of source

The overall similarity in REE and trace element compositions (Fig. 7a and b) provides strong evidence that all three igneous suites originated from a similar source. Characterization of this source is not straightforward, as petrogenetic modeling of these magma compositions must account for possible effects of being generated and emplaced in an intracontinental tectonic setting. Inspection of chondrite-normalized REE patterns for the samples (Fig. 7a) reveals both the enrichment of all REEs, as well as the pronounced enrichment of the incompatible LREEs. This highly enriched LREE pattern for mafic rocks cannot originate solely from a depleted mantle source (i.e. source of mid-ocean ridge basalt; MORB), even from very small degree partial melts. This assessment is supported by primitive mantle-normalized trace element patterns of these rocks

Table 3
Sm–Nd isotopic data for selected MGIR.

Sample	Sm (ppm)	Nd (ppm)	$^{147}Sm/^{144}Nd$	$^{143}Nd/^{144}Nd$	2σ unc.	ε_{Nd} (1818 Ma)	T_{DM}[a] (Ma)
RM-6v	9.45	64.35	0.08876	0.511096	0.000013	−4.9	2499
RM-8v	13.23	94.59	0.08456	0.510974	0.000007	−6.3	2561
RM-10v	24.96	186.54	0.08085	0.511021	0.000012	−4.5	2437
RM-17d	13.81	99.66	0.08372	0.511076	0.000014	−4.1	2427
RM-18d	20.59	161.07	0.07727	0.510901	0.000008	−6.0	2506
RM-26d	7.04	46.03	0.09251	0.511104	0.000027	−5.6	2566
RM-29s	27.41	200.41	0.08266	0.511022	0.000015	−4.9	2469
RM-31s	14.84	104.44	0.08588	0.511074	0.000012	−4.7	2469

Last character of sample name denotes analysis of volcanic (v), sill (s), or dike (d). unc. = uncertainty.
[a] model of Goldstein et al. (1984).

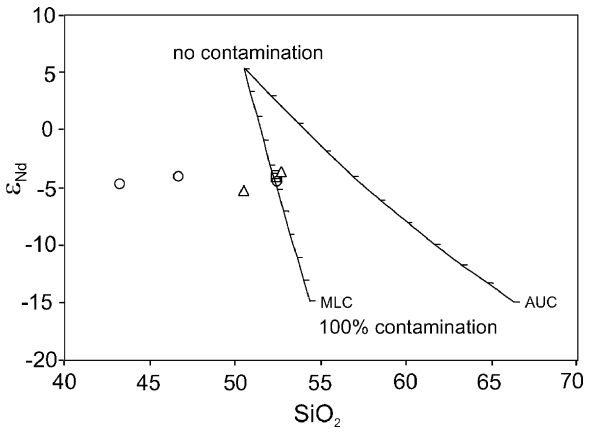


Fig. 10. Mixing model showing plots of ε_{Nd} vs. total silica mixing between normal MORB at 1818 Ma and both hypothetical 'mafic lower crustal' (MLC) and 'andesitic upper crustal' (AUC) end members to assess the possible role of crustal assimilation for the Martin Group magmas.

(Fig. 7b). Depleted MORB mantle (DMM) is characterized by relatively low Th contents, with Th/Ta_N ratios of ~0.4 (Workman and Hart, 2005). This overall trace element pattern is transferred upon partial melting of this source to the resulting MORB (N-MORB, Fig. 7b), albeit at higher concentrations. This contrasts with the measured compositions of the volcanic rocks, sills and dikes, which collectively have Th/Ta_N ratios ranging from 2.9 to 5.4. Such a signature would be unlikely to result from any degree of melting of a DMM due the nominal difference in the incompatibilities of Th and Ta.

In addition to the enrichment of incompatible elements, the MGIR extended trace element patterns have comparatively low Ta and Nb contents, and slight relative depletions of Zr and Ti with respect to neighbouring elements. This combined enrichment of incompatible elements and relative depletion of HFSE in alkaline magmas has been interpreted to reflect inheritance by the magma of a subduction signature (e.g. Luhr, 1997; Cousens et al., 2001), produced due to the preferential retention of HFSE in a subducting slab during dehydration reactions. Specifically, metasomatic fluids and/or melts derived from the slab infiltrate the overlying mantle, thereby instilling the depleted HFSE signature to this mantle and any rocks derived from melting of it.

The observed REE and extended trace element patterns of the MGIR therefore require that the magma was either generated in a reservoir characterized by high incompatible element concentrations, or experienced significant amounts of contamination (via assimilation–fractional crystallization [AFC] processes) by crust with high incompatible element concentrations and HFSE negative anomalies. Considering the latter possibility, the most plausible scenario to result in the observed MGIR compositions is the assimilation of older, intermediate to felsic, subduction-derived crust (e.g. arc-related rocks derived from melting of the mantle wedge) through AFC processes in a magma chamber. With the available geochemical and isotopic data for the MGIR, it is difficult to unambiguously demonstrate the importance of assimilation and fractionation processes in the generation and/or emplacement of these magmas. However, some general observations can be made on the basis of a simple two-component (ε_{Nd} and wt.% SiO_2) mixing model, which can help to assess whether crustal contamination alone can produce the required magma compositions (Fig. 10). By mixing a melt from 1818 Ma 'depleted mantle' (Goldstein et al., 1984), approximating contemporaneous normal MORB (Presnall and Hoover, 1987), with a hypothetical andesitic contaminant (60 wt.% SiO_2) having a prolonged crustal residence time ($\varepsilon_{Nd} = -15$ @ 1818 Ma), it is clear that only a magma too silicic to match the required compositions can be produced. In contrast, repeating this exercise using hypothetical mafic lower crust (54.5 wt.% SiO_2; Taylor and McLennan, 1985) as an end member contaminant can produce the required Nd isotopic and SiO_2 compositions by approximately 40–50% contamination of a MORB melt with such crust (Fig. 10). However, despite agreement in the resulting SiO_2 and ε_{Nd} of this mixture for some samples, the deficiency in Th and LREE elements in the lower crust (Fig. 7a) relative to the MGIR eliminate this mixture as a probable origin for these rocks.

Further observations from the major and trace element compositions strengthen the argument against crustal contamination being a significant process in the generation of these enriched magma compositions. The MGIR samples displaying the most enriched trace element patterns have relatively low silica contents and lack negative Eu anomalies, contrary to what would be expected considering the LREE enriched, plagioclase-rich nature of felsic continental crust. The absence of crustal xenoliths in the majority of the MGIR further indicates that crustal contamination was likely insignificant in generating the observed compositions, as does the fact that this highly enriched signature characterizes the sample with the lowest silica content (i.e. that most likely to have been influenced by AFC processes; volcanic sample RM-8, 43% SiO_2). Although these observations do not preclude the possibility that there was modification of the primary melt compositions by assimilated continental crust during fractional crystallization, they suggest that a depleted mantle-derived magma contaminated by enriched continental crust was unlikely to be the source of the MGIR.

An alternative explanation for the generation of this chemical signature is that these magmas were derived from subcontinental lithospheric mantle that had previously experienced chemical modification by metasomatic and/or magmatic processes (e.g. Pearce, 1983; Hawkesworth et al., 1984; Menzies et al., 1987; Kempton et al., 1991). One way this possibility can be explored further is by inspection of the relative proportions of trace elements in the MGIR. Pearce (1983) has shown that some magmas generated at active continental margins contain detectable input from subcontinental lithosphere and, furthermore, that magmas generated in continental settings can contain contributions from at least three distinct components (i.e. 'mantle', 'subduction', and 'within-plate'). In this model, the mantle component represents depleted mantle, similar in composition to that melted during the generation of tholeiitic MORB, and is distinguished by the concentrations of compatible trace elements that reside within refractory mineral phases during partial melting at depth (e.g. Y, Yb). The strong enrichment of incompatible elements with a higher potential for mobility in an aqueous fluid (e.g. Sr, K, Rb, Ba) indicates a 'subduction zone' component, referring to subduction-derived fluid transfer into the subcontinental mantle lithosphere. Enrichment of elements not accommodated by either the mantle or subduction zone components (e.g. HFSE) indicate derivation from the 'within-plate' component. The origin of this incompatible element enriched component is not readily defined due to the uncertainty of the processes that might affect the subcontinental lithosphere, but is similar in composition to that observed for ocean island basalts (e.g. Hawaiian basalts; Pearce, 1983).

The procedure of Pearce (1983) is applied here to assess whether similar components are recognizable in the MGIR. In doing so, we used the sample that is the least enriched in incompatible HFSE (dike sample RM-26) as a representative, though all dike, volcanic, and sill samples yield similar results. All three enrichment components are identifiable in the MGIR (Fig. 7c), indicating a complex enrichment history for their presumed subcontinental mantle lithospheric source. If the effects of all enrichment are removed, it is apparent from the Y and Yb concentrations that the there is an original component of depleted mantle that resulted in an ele-

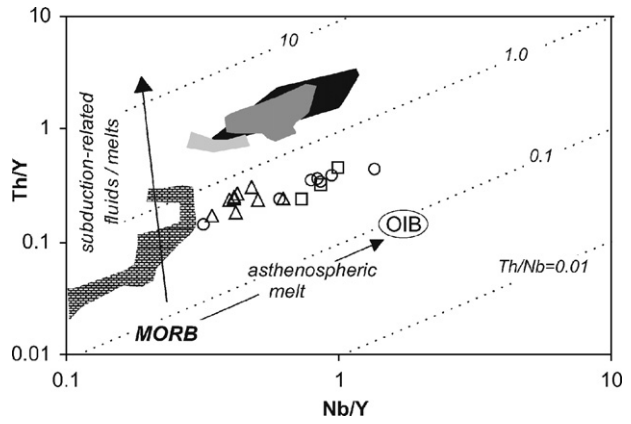


Fig. 11. Th/Y vs. Nb/Y variation diagram for the Martin Group igneous suites (after Kempton et al., 1991; Seyitoğlu et al., 1997). The compositional field for ocean island basalt ('OIB') is from Sun and McDonough (1989). Also shown for comparison are the fields for basalts of the Murmac Bay Group (Hartlaub et al., 2004, stipple pattern) and Beaverlodge Domain granitoid rocks of variable age (3.0 Ga = black, 2.3 Ga = dark grey, 1.9 Ga = light grey).

mental contribution with a relatively flat distribution at ~0.65× MORB. This component is perhaps representative of the starting composition of the subcontinental lithospheric mantle following the earlier extraction of incompatible element rich melts. This depleted mantle subsequently underwent a subduction-related enrichment, which contributed the majority of K, Ba, Th, and Ce to the source. Although the relative timing is unclear, an additional component characteristic of within-plate magmas also contributed to the Martin Group magmas that accounts for the majority of the budget of HFSE.

Further discrimination is required to assess the possible origin of the components detected in the MG magmas. The use of a bivariate ratio plot of Th/Y vs. Nb/Y (or Th/Yb vs. Ta/Yb; Pearce, 1983) has been used to help determine the origin of such contributions in the source of other alkaline magmas (e.g. Kempton et al., 1991; Seyitoğlu et al., 1997), and was utilized here to elucidate processes that might have affected the subcontinental lithospheric mantle source of the MGIR (Fig. 11). Two separate enrichment vectors characterize this plot, such that increasing Th/Y (or Th/Nb) and Nb/Y ratios signify enrichment of an initially depleted mantle source by subduction-related crustal input and by small degree asthenospheric melts, respectively. The relative Th and Nb compositions of the MGIR (Fig. 11) suggest that, by the time of emplacement of the MG magmas at ca. 1818 Ma, the subcontinental mantle source had interacted with small degree asthenospheric melts and been chemically enriched by components derived from a subducted slab. We equate the asthenospheric component, which has compositional characteristics similar to ocean island basalt (OIB; Fig. 11), to the HFSE enriched 'within-plate' component identified in Fig. 7c. The time at which this component was incorporated into the MG magma source is unknown. It is possible that this was already part of the magma source when tapped at ca. 1818 Ma, perhaps representing asthenospheric melts that had infiltrated the overlying subcontinental lithospheric mantle during an earlier melting event. Alternatively, E-W directed shortening of the western Churchill Province at 1818 Ma (Ashton et al., in press) resulted in the formation of intracontinental basins and could have produced lithospheric thinning and generation of partial melts via adiabatic decompression (Cousens et al., 2001). It is therefore possible that localized crustal extension resulting in opening of the Martin basin and related magmatism at ca. 1818 Ma could have coincided with lithospheric thinning, initiation of asthenospheric melting, and interaction of the melts with the overlying lithospheric mantle.

For characterization of the subduction-related component, it is noteworthy that mantle-derived melts with Th/Nb ratios elevated above those of MORB, such as those for the MGIR (Th/Nb = 0.32–0.63), can be attributed to a source enriched by continental crust, postulated to represent fluids and/or small degree melts derived from subducted sediment (Rogers et al., 1985); Kempton et al., 1991). Furthermore, Kempton et al. (1991) suggest that both slab-derived metasomatism and melting of slab and/or slab sediments can occur during a subduction event, and that these subduction-related enrichment processes are not required to add large amounts of recycled (crustal) material to significantly change the trace element composition of the produced melt, as would be necessary during crustal assimilation in a magma chamber.

Based on the major and trace element compositions of the MGIR, we propose that these magmas were initially derived from melting of subcontinental lithospheric mantle that was originally chemically depleted and that subsequently experienced chemical modification from asthenospheric mantle melts as well as subduction-derived components, including fluids and/or melts of subducted sediment. Although the significance is currently unclear, another interesting aspect of the MGIR compositions shown in Fig. 11 is the apparent continuum of Th and Nb compositions defined by the MGIR with those of the older (2330–1930 Ma) Murmac Bay Group basalts. If these suites were ultimately derived from the same mantle source, the common trajectory of the line defined by the Th/Nb ratios of these two suites, parallel to that defining increasing asthenospheric input (i.e. higher Nb/Y ratios) could indicate that some of the mantle source enrichment occurred after the extraction of the Murmac Bay Group mafic melts. Furthermore, in accordance with our previous interpretation, Fig. 11 supports the inference that assimilation of upper crustal material was unlikely to have been a major process in the generation of these magmas, since older Beaverlodge Domain granitoid rocks, which collectively constitute a significant proportion of Beaverlodge Domain upper crust (Hartlaub, 2004), have significantly higher Th/Nb ratios than the MGIR. Overall, this interpretation is consistent with the view that the subcontinental mantle acts as a repository for fluids and melts of varying compositions and derived from different processes. Such compositional variation is consistent with lithologic sections of global examples of subcontinental mantle lithosphere constructed from xenocryst/xenolith compositions (Griffin et al., 1999; Poudjom Djomani et al., 2001; O'Reilly et al., 2001).

6.3. Intersuite compositional and temporal variations

The overall compositional similarities of the three MGIR suites (sills, volcanic rocks, dikes), including the extreme LREE enrichment and relative HFSE depletions, suggest that the individual magma suites were ultimately derived from a similar source reservoir. Furthermore, field relationships between the Uranium City mafic dikes and the basal units of the MG (see Fig. 3) indicate that the three MGIR suites were emplaced over a relatively short time interval. In detail, however, slight but discernable compositional variations do exist (see Figs. 6, 7 and 9) and are sufficient to suggest slightly different evolutionary histories for each suite. The Uranium City dikes are consistently subalkaline (i.e. tholeiitic), are slightly deficient in incompatible trace elements relative to the other two suites, and display remarkable intrasuite compositional uniformity throughout the area (>3500 km^2) of currently known exposures. The sills and the volcanic rocks are slightly more alkaline, have generally elevated incompatible trace element abundances (corresponding to increasing modal apatite abundances), and display much more intrasuite compositional variation between individual intrusions/flows. This intersuite compositional variation might reflect: (i) differential effects of AFC processes in the magmas that were superimposed on the source signature, (ii) different degrees of

partial melting of the same source, and/or, (iii) slight, localized compositional variation in the source region. Whatever the cause, these geochemical differences collectively indicate that, although likely having a close genetic relation, none of the sampled dikes directly fed magma to the exposed intrabasinal sills and volcanic rocks of the MG. The relative geochemical character of the three suites suggests that the dikes might represent a more primitive magma from which the sills and volcanic rocks were derived.

The new U–Pb baddeleyite date presented here indicates that Uranium City dike intrusion occurred, at least in part, at ca. 1818 Ma. Given that these dikes are observed only to cut the basal unit (Beaverlodge Formation) of the MG, we interpret this to represent the timing of the first pulse of lithospheric mantle-derived MG magmatism, though it is possible that dike emplacement was an extended event that lasted until the cessation of Martin Group volcanism. Using the model of Ashton et al. (2001) as a structural framework for basin development, we propose these dikes infiltrated faults and fractures that originated prior to or during early D4 deformation in the Beaverlodge Domain. Early dike emplacement was succeeded by a brief period of magmatic quiescence, concurrent with deposition of sediments that now comprise the Boulder Conglomerate and Middle Sandstone Members of the Gillies Channel Formation (Fig. 3). Magmatic activity then renewed with the intermittent emplacement of the MG sills and volcanic flows during deposition of the Upper Sandstone Member. This later magmatism possibly occurred because of reactivated displacement along major faults (i.e. the Black Bay and St. Louis faults) during continued opening and subsidence of the basin, consequently retapping the magma chamber and providing new conduits through which magma could ascend to (or near to) surface.

6.4. Regional considerations and implications for Sm–Nd isotopic compositions

Previously, correlations have been suggested between the Martin Group and the Baker Lake Group (Dubawnt SuperGroup), the latter of which is exposed over a large area approximately 500 km to the northeast of Uranium City (Donaldson, 1968; Peterson et al., 2002). The Baker Lake Group contains alkaline volcanic rocks, specifically incompatible element enriched, minettes and lamproites of the Christopher Island Formation (CIF; Peterson et al., 1994; Rainbird et al., 2006). This volcanism was coincident with emplacement of a regionally extensive lamprophyre dyke suite (Fig. 1) and a suite of granitic plutons (the 1845–1795 Ma 'Hudson' granites; van Breeman et al., 2005). Petrogenetic models for the most mafic CIF magmas favor enrichment of the subcontinental lithosphere by subduction-related fluids or melts (Peterson et al., 1994, 2002; Cousens et al., 2001). A source model proposing mixing of mantle melts and crust was also previously considered (Peterson et al., 2002) but subsequently, as proposed here for the MGIR, it was determined that crustal contamination of mantle-derived melts was not a viable process for generating the observed CIF magma compositions (Peterson, 2006).

The existing 1830–1810 Ma age constraints for the CIF (Rainbird et al., 2006) are contemporaneous with the new 1818 Ma age presented here for a Uranium City mafic dyke, which is interpreted to have a close temporal relation to Martin Group deposition. This generally corroborates the Martin Group–Baker Lake Group correlation in terms of the timing of sediment deposition and related magmatism, and indicates that basin formation occurred as a result of a common stress regime at this time. The geochemical data presented here reveal the alkaline tendencies of the MGIR and their enrichment in incompatible elements with relative depletions in high field strength elements, similar to that observed for the CIF magmas (Cousens et al., 2001; Peterson, 2006). This does not necessarily require derivation of the MGIR and CIF magmas from a common source, however, as some slight differences between the MGIR and CIF are also noted. These include the more basalt-like major element composition of the MGIR relative to the ultrapotassic CIF, and the paucity of contemporaneous granitic magmatism in the immediate vicinity of the Martin Group, with the nearest identified pluton of this age situated ~70 km to the east in the Train Lake Domain (Card, 2001; Ashton et al., in press). Based on these criteria, Ashton et al. (in press) suggest that MGIR are more similar to the Sparrow dykes in the western Rae craton (Fig. 1), whereas the CIF magmas compare more favorably to a suite of alkali feldspar and phlogopite bearing lamprophyre dikes exposed in the Train Lake and eastern Beaverlodge domains, immediately west of the Black Bay Fault and of the present study area (Fig. 1). These spatial relationships between compositionally distinctive, ca. 1820 Ma dike suites might suggest a subtle compositional variation in the Rae subcontinental lithosphere, with a boundary approximated by the position of the Black Bay Fault.

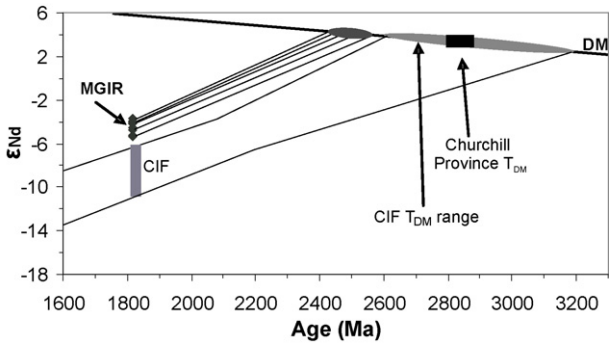


Fig. 12. Diagram of ε_{Nd} vs. time showing the Nd isotopic evolution of the magma source for the Martin Group igneous rocks relative to depleted mantle (Goldstein et al., 1984). Shown for comparison are Nd isotopic data for Proterozoic igneous rocks of the Christopher Island Formation (CIF, Cousens et al., 2001) and average Nd model ages for Saskatchewan Churchill Province (McCulloch and Wasserburg, 1978).

There are currently no published Sm–Nd isotopic data for the Sparrow dikes or the eastern Beaverlodge lamprophyre dikes; thus comparison between the MGIR and CIF Sm–Nd isotopic data provides a useful means of assessing regional compositional variation in the Rae craton sublithospheric mantle. Subduction-related mantle enrichment resulted in the eventual transfer of evolved Nd isotopic compositions to the CIF magmas ($\varepsilon_{Nd} = -6$ to -11 at 1830 Ma) and, correspondingly, ancient Sm–Nd depleted mantle model ages (T_{DM}) that cluster at about 2.8 Ga (Cousens et al., 2001; Peterson, 2006), similar to that of average Rae craton crust (Fig. 12). This isotopic signature has been interpreted to have originated during an Archean subduction event (Cousens et al., 2001) or from Archean subducted sediment during a Proterozoic subduction event (Peterson et al., 1994). Recent recognition of evidence for 1.9 Ga collisional orogenesis along the boundary between the Rae and Hearne cratons ('Snowbird Tectonic Zone'; Berman et al., 2007) could, however, point to the latter as the more likely model. The MGIR exhibit a similarly enriched mantle source, albeit with a slightly less negative range of ε_{Nd} values of between -4 and -6 (Table 3) and, owing to a parallel evolutionary trajectory with the CIF (Fig. 12), slightly 'younger' T_{DM} values that cluster at about 2.5 Ga. As suggested by MGIR trace element compositions and consistent with that interpreted for the CIF magma source, the presence of this enriched Nd component suggests that a subduction-related process also affected the mantle lithosphere source region of these magmas. However, although both suites clearly carry this signature, the differences between the composition and lithological associations of the CIF and MGIR are sufficient to allow for the possibility that this enrichment might not have resulted from the same subduction event. As discussed by Ashton et al. (in press), multiple

Paleoproterozoic orogenic episodes could have affected the mantle to this part of the Rae craton, either cumulatively or in isolation. This enrichment could have occurred during the ca. 1.95 Ga Taltson-Thelon Orogeny to the west and/or the ca. 1.8 Ga Trans-Hudson Orogeny to the southeast (Fig. 1), or possibly even during orogenesis along the western Rae margin at ca 2.3 Ga (Arrowsmith Orogeny'; Berman et al., 2005; Hartlaub et al., 2007) or along the Snowbird Tectonic Zone at ca. 1.9 Ga (Berman et al., 2007). Alternatively, these differences could reflect spatial variations as a result of a single subduction event, such that the effects of this subduction on the Rae subcontinental lithosphere are more prevalent in the CIF/lampropyhre source region to the east of the Black Bay Fault.

The observed variation between the Sm–Nd isotopic compositions of the MGIR and CIF, together with the uncertainty in the event(s) that resulted in enrichment of their source mantle, renders these data unsuitable for interpreting the timing of mantle enrichment events. The evolved isotopic signature of these magmas most likely reflects the original composition of material added to the sublithospheric mantle (i.e. the subducted material) as opposed to the timing of subduction or the mantle enrichment event. Furthermore, if our interpretation that depleted mantle and an asthenospheric melt also contributed to the MGIR source is correct, the resulting Sm–Nd isotopic signature could reflect a mixture of several components. For example, any contributions from asthenosphere or depleted mantle would cause an upward shift of the pre-1818 Ma mantle to less negative ε_{Nd} values. Consequently, the isotopic evolutionary trajectories depicted in Fig. 12 are interpreted to represent an average composition of all components present in the source region, and probably do not uniquely define the timing of the mantle enrichment event(s) or any other individual geological event.

7. Conclusions

New U–Pb geochronology, Sm–Nd isotopic data, and major and trace element geochemical data permit interpretation of the origin and emplacement history of a mafic dike suite located in the Uranium City area, and comparison with that of mafic volcanic rocks and sills within the nearby Paleoproterozoic Martin Basin. U–Pb isotopic analysis of baddeleyite from one dike indicates an emplacement age of 1818 ± 4 Ma for this suite, consistent with emplacement in an intraplate tectonic setting at that time. Crosscutting relationships of the dikes with rocks of the Martin Group are consistent with the emplacement of all three suites within a relatively short period of time. This relationship is generally confirmed by major and trace element compositions that reveal derivation of the three suites from a similar source, but with sufficient variation to indicate slight differences in timing of emplacement and/or magmatic histories. Elemental and Sm–Nd isotopic compositions of the rocks argue against the modification of depleted mantle melts via AFC processes as the main magma source, and instead reflect derivation of all three suites from enriched subcontinental mantle lithosphere. Initially having a depleted composition mantle, this lithospheric mantle experienced chemical contributions from multiple components prior to magma generation, including metasomatic fluids and/or melts derived from an earlier subduction event, and small degree melts that are similar in composition to ocean island basalt. The timing of the events resulting in this enrichment is unknown. This magmatism possibly occurred because of adiabatic melting related to localized lithospheric thinning of the western Rae protocraton.

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